QUALCOMM PM6150 PMIC POWER RAILS SCHEMATIC

Salen Mobile Guru Hardware Blueprint | Sheet 1 of 3

POWER DISTRIBUTION & INPUT STAGE

Primary VPH_PWR and Battery Fuel Gauge Interconnects

VBAT Input: Pin A1, A2, A3 (3.7V - 4.35V nominal DC)

VPH_PWR Switching Output: Coil L1001 (1.0uH 4.5A) -> Diode Mode to GND: 420mV

Cold Resistance on VPH_PWR to GND: > 15K Ohms (healthy board)

Leakage Threshold: Any current > 0.005A on standby indicates secondary line leak

Charge Pump Boost Line: Pin C4 to C1045 (Diode Mode: 510mV, Boost: 5.4V during display ON)

SMB1390 Fast Charge Interconnect: I2C address 0x18 (SDA: 1.8V, SCL: 1.8V)

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BUCK CONVERTER OUTPUT SEQUENCE

Secondary Regulated Voltage Supplies for Application Processor

BUCK S1A (VDD_CORE): 0.75V - 0.95V @ 6.0A | Coil L1002 | Diode: 180mV

BUCK S2A (VDD_MODEM): 0.65V - 0.85V @ 4.0A | Coil L1003 | Diode: 210mV

BUCK S3A (VDD_RAM LPDDR4X): 1.125V @ 3.5A | Coil L1004 | Diode: 260mV

BUCK S4A (VDD_MEM UFS): 1.80V @ 2.5A | Coil L1005 | Diode: 410mV

BUCK S5A (VREG_BOB): 3.30V System Boost | Coil L1006 | Diode: 480mV

Diagnostic Rule: If S1A or S3A shows < 15mV in diode mode, CPU core is shorted

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LDO VOLTAGE RAILS & SLEEP OSCILLATOR

Low-Dropout Regulators and Clock Management

LDO 1A (VREG_L1A_1P0): 1.0V clean analog rail for PLL | Diode: 390mV

LDO 5A (VREG_L5A_1P8): 1.8V digital sensor & pull-up supply | Diode: 450mV

LDO 6A (VREG_L6A_2P8): 2.8V Touch IC & Camera analog supply | Diode: 520mV

32.768 KHz Sleep Clock (XO_CLK): Pin E5 -> Amplitude: 1.2V P-P

PMIC Reset Line (PS_HOLD): High (1.8V) from CPU to PMIC within 120ms of boot

Power Key (PWR_KEY_N): Pulled high to 1.8V by internal 100K resistor

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